Introduction & Using R

Motivations for mathmatical modelling → Inform, Curiosity, Predictions, Control

Model = simplified description (mathmatical) of system / process to assist with calcs / prediction

Statistic Model

- Look for patterns in exsisting data
- Simple equations (linear models, regression)
- Lot of data needed
- Suitable for interpolation (between points / missing)
- NOT suitable for prediction!

✡ Mechanistic Model

- Use underlying understanding
- complex equations
- Minimal data
- Suitable for interpolation + prediction

Types of mechanistic modelling

Deterministic

- Entirely predictable
- Fixed set of initial conditions
- Outcome always the same
- output represents 'mean' outcome
- Run model once (Computationally cheap)
- Hard to quantify uncertainty
- More appropriate for large populations

Stochastic

- Random variable inputs
- Random outputs, different every time
- Run many times (computationally expensive)
- Can quantify uncertainty
- More appropriate for small populations

Spread of infectious disease has many variables

Individual based modelling is computationally expensive & complex

Homogeneous mixing - everyone is just as likely to meet

''' - backstroke {} ''' - code chunk

echo = False - don't print code

library (tidyverse) = similar to Matplot Lib

- comment

<- assign data to variable

|> or %>% = runs function

Read_csv = not . but _